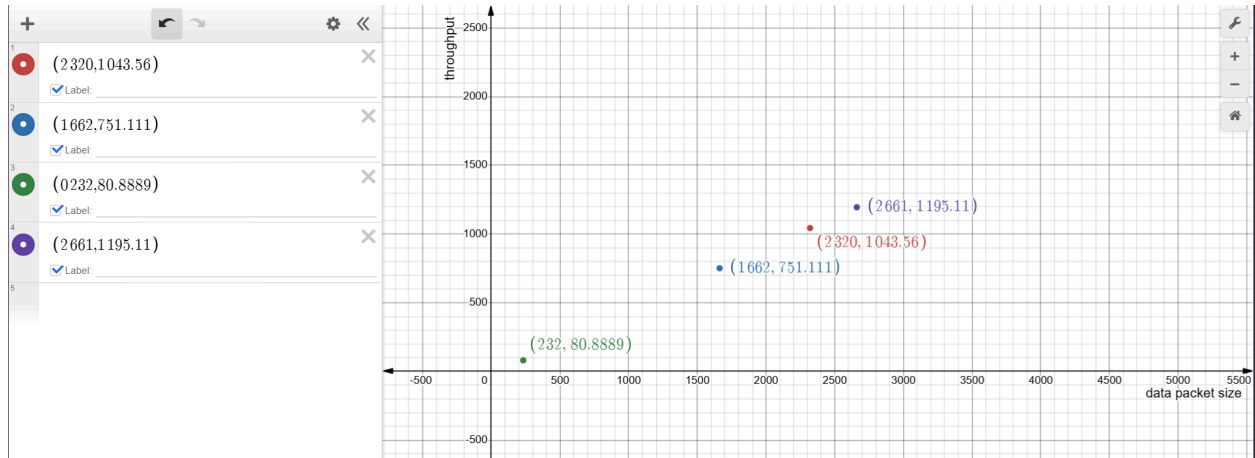


```
student@administrator-HP-280-Pro-G6-Microtower-PC: ~/Downloads/ns-a...
54 | echoClient.SetAttribute "PacketSize", UIntegerValue(2320);
ninja: build stopped: subcommand failed.
student@administrator-HP-280-Pro-G6-Microtower-PC: ~/Downloads/ns-allinone-3.43/n
s-3.43$ ./ns3 run scratch/first.cc
[0/2] Re-checking globbed directories...
[2/3] Linking CXX executable /home/stu...43/build/scratch/ns3.43-first-default
At time +2s client sent 2320 bytes to 10.1.1.2 port 9
At time +2.0058s server received 2320 bytes from 10.1.1.1 port 49153
At time +2.0058s server sent 2320 bytes to 10.1.1.1 port 49153
At time +2.01159s client received 2320 bytes from 10.1.1.2 port 9
FlowID: 1 (UDP 10.1.1.1/49153 --> 10.1.1.2/9)
Tx Bytes: 2348
Rx Bytes: 2348
Mean Delay: 0.0057952 s
Throughput: 1043.56 bps
FlowID: 2 (UDP 10.1.1.2/9 --> 10.1.1.1/49153)
Tx Bytes: 2348
Rx Bytes: 2348
Mean Delay: 0.0057952 s
Throughput: 1043.56 bps
student@administrator-HP-280-Pro-G6-Microtower-PC:~/Downloads/ns-allinone-3.43/n
s-3.43$
:Stop(Seconds(20.0));
:Run();
```

```
student@administrator-HP-280-Pro-G6-Microtower-PC: ~/Downloads/ns-a...
Tx Bytes: 2348
Rx Bytes: 2348
Mean Delay: 0.0057952 s
Throughput: 1043.56 bps
student@administrator-HP-280-Pro-G6-Microtower-PC: ~/Downloads/ns-allinone-3.43/n
s-3.43$ ./ns3 run scratch/first.cc
[0/2] Re-checking globbed directories...
[2/3] Linking CXX executable /home/stu...43/build/scratch/ns3.43-first-default
At time +2s client sent 1662 bytes to 10.1.1.2 port 9
At time +2.00474s server received 1662 bytes from 10.1.1.1 port 49153
At time +2.00474s server sent 1662 bytes to 10.1.1.1 port 49153
At time +2.00948s client received 1662 bytes from 10.1.1.2 port 9
FlowID: 1 (UDP 10.1.1.1/49153 --> 10.1.1.2/9)
Tx Bytes: 1690
Rx Bytes: 1690
Mean Delay: 0.0047424 s
Throughput: 751.111 bps
FlowID: 2 (UDP 10.1.1.2/9 --> 10.1.1.1/49153)
Tx Bytes: 1690
Rx Bytes: 1690
Mean Delay: 0.0047424 s
Throughput: 751.111 bps
student@administrator-HP-280-Pro-G6-Microtower-PC: ~/Downloads/ns-allinone-3.43/n
s-3.43$
Sim
:::Stop(Seconds(20.0));
:::Run();
```

```
first.cc
student@administrator-HP-280-Pro-G6-Microtower-PC: ~/Downloads/ns-a...
Interface Tx Bytes: 260
           Rx Bytes: 260
choServer  Mean Delay: 0.0024192 s
           Throughput: 115.556 bps
licationCo student@administrator-HP-280-Pro-G6-Microtower-PC: ~/Downloads/ns-allinone-3.43/n
erApps.St $ ./ns3 run scratch/first.cc
erApps.St [0/2] Re-checking globbed directories...
erApps.St [2/3] Linking CXX executable /home/stu...43/build/scratch/ns3.43-first-default
           At time +2s client sent 154 bytes to 10.1.1.2 port 9
choClient At time +2.00229s server received 154 bytes from 10.1.1.1 port 49153
client.Se At time +2.00229s server sent 154 bytes to 10.1.1.1 port 49153
client.Se At time +2.00459s client received 154 bytes from 10.1.1.2 port 9
client.Se FlowID: 1 (UDP 10.1.1.1/49153 --> 10.1.1.2/9)
           Tx Bytes: 182
           Rx Bytes: 182
htApps.St Mean Delay: 0.0022944 s
htApps.St Throughput: 80.8889 bps
           FlowID: 2 (UDP 10.1.1.2/9 --> 10.1.1.1/49153)
. Setup F Tx Bytes: 182
MonitorHe Rx Bytes: 182
FlowMonit Mean Delay: 0.0022944 s
           Throughput: 80.8889 bps
. Run Sim student@administrator-HP-280-Pro-G6-Microtower-PC: ~/Downloads/ns-allinone-3.43/n
s-3.43$
lator::Stop(Seconds(20.0));
lator::Run();
```

```
student@administrator-HP-280-Pro-G6-Microtower-PC: ~/Downloads/ns-a...  
Tx Bytes: 182  
Rx Bytes: 182  
Mean Delay: 0.0022944 s  
Throughput: 80.8889 bps  
student@administrator-HP-280-Pro-G6-Microtower-PC: ~/Downloads/ns-allinone-3.43/n  
s-3.43$ ./ns3 run scratch/first.cc  
[0/2] Re-checking globbed directories...  
[2/3] Linking CXX executable /home/stu....43/build/scratch/ns3.43-first-default  
At time +2s client sent 2661 bytes to 10.1.1.2 port 9  
At time +2.00634s server received 2661 bytes from 10.1.1.1 port 49153  
At time +2.00634s server sent 2661 bytes to 10.1.1.1 port 49153  
At time +2.01268s client received 2661 bytes from 10.1.1.2 port 9  
FlowID: 1 (UDP 10.1.1.1/49153 --> 10.1.1.2/9)  
Tx Bytes: 2689  
Rx Bytes: 2689  
Mean Delay: 0.0063408 s  
Throughput: 1195.11 bps  
FlowID: 2 (UDP 10.1.1.2/9 --> 10.1.1.1/49153)  
Tx Bytes: 2689  
Rx Bytes: 2689  
Mean Delay: 0.0063408 s  
Throughput: 1195.11 bps  
student@administrator-HP-280-Pro-G6-Microtower-PC: ~/Downloads/ns-allinone-3.43/n  
s-3.43$  
Stop(Seconds(20.0));  
Run();
```



The graph demonstrates a clear positive correlation between the two variables. As the data packet size increases (moving right along the x-axis), the throughput also steadily increases (moving up the y-axis). This is a common occurrence in networking, as transmitting larger packets reduces the ratio of header overhead to actual payload data, thereby increasing the effective throughput.